

Evidence on the Human Health Effects of Low-Level Methylmercury Exposure

Background: Methylmercury (MeHg) is a known neuro-toxicant. Emerging evidence indicates it may have adverse effects on the neuro-logic and other body systems at common low levels of exposure. Impacts of MeHg exposure could vary by individual susceptibility or be confounded by bene-ficial nutrients in fish containing MeHg. Despite its global relevance, synthesis of the available literature on low-level MeHg exposure has been limited. Objectives: We undertook a synthesis of the current knowledge on the human health effects of low-level MeHg exposure to provide a basis for future research efforts, risk assessment, and exposure remediation policies worldwide. Data sources and extraction: We reviewed the published literature for original human epidemio-logic research articles that reported a direct biomarker of mercury exposure. To focus on high-quality studies and those specifically on low mercury exposure, we excluded case series, as well as studies of populations with unusually high fish consumption (e.g., the Seychelles), marine mammal consumption (e.g., the Faroe Islands, circumpolar, and other indigenous populations), or consumption of highly contaminated fish (e.g., gold-mining regions in the Amazon). Data synthesis: Recent evidence raises the possibility of effects of low-level MeHg exposure on fetal growth among susceptible subgroups and on infant growth in the first 2 years of life. Low-level effects of MeHg on neuro-logic outcomes may differ by age, sex, and timing of exposure. No clear pattern has been observed for cardio-vascular disease (CVD) risk across populations or for specific CVD end points. For the few studies evaluating immunologic effects associated with MeHg, results have been inconsistent. Conclusions: Studies targeted at identifying potential mechanisms of low-level MeHg effects and characterizing individual susceptibility, sexual dimorphism, and non-linearity in dose response would help guide future prevention, policy, and regulatory efforts surrounding MeHg exposure.